

TrES-3b

R-0319

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_170519 · created 2026-07-18T01:27:57+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2457892.87 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.156 +/- 0.016
Transit depth (Rp/Rs)^2	2.42 +/- 0.51 [%]
Orbital Inclination (inc)	83.88 +/- 1.34
Airmass coefficient 1 (a1)	1.299 +/- 0.015
Airmass coefficient 2 (a2)	-0.248 +/- 0.031
Scatter in the residuals of the lightcurve fit is	1.14 %
Best Comparison Star	None
Optimal Aperture	3.69
Optimal Annulus	13.82
Transit Duration (day)	0.0677 +/- 0.008

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.156 ± 0.016 vs 0.16309 (deviation 0.3σ)
Duration fit vs published	0.0677 d vs 0.0567 d (deviation 0.92σ)
Mid-time O–C	2.4 min
Depth SNR	6.5
Residual scatter	1.14 %

Light curve

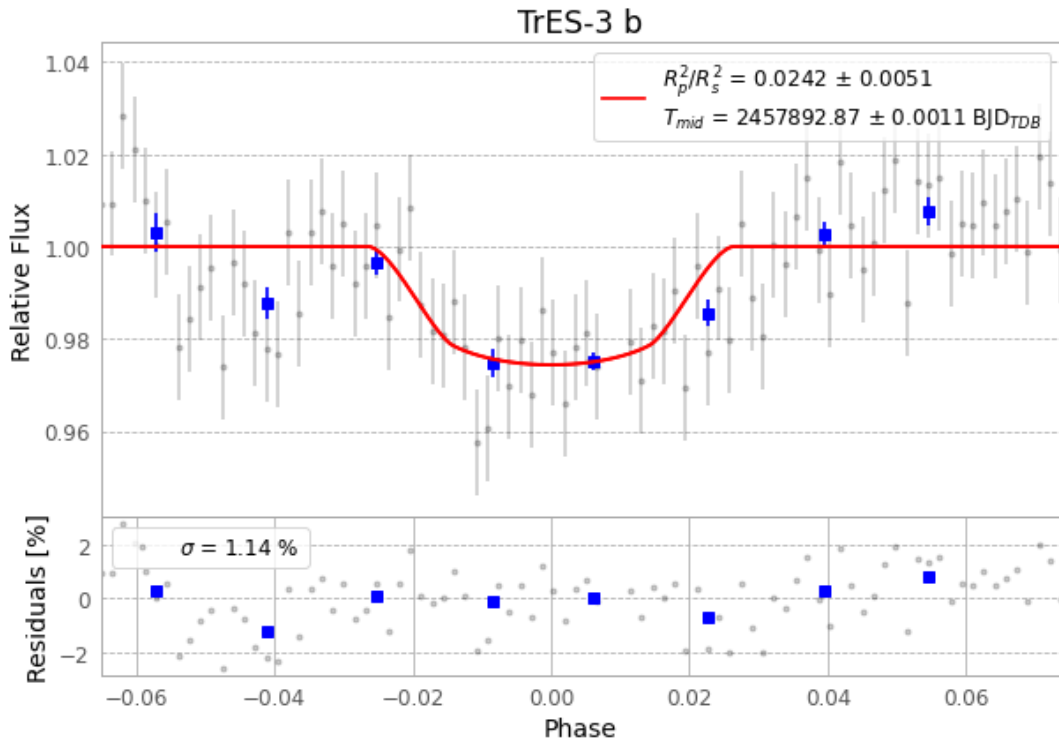


Figure 1 — FinalLightCurve_TrES-3 b_2017-05-18.png (SHA-256 daf8fe450b0dbc...

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [313, 246], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 588c8ca51cc61c60...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1bbef5c

Data provenance

86 FITS frames (56 MB), TRES-3170519064512.FITS → TRES-3170519110612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-170519.log (9b6a419d11f3...), FinalLightCurve_TrES-3 b_2017-05-18.png (daf8fe450b0d...), FinalLightCurve_TrES-3 b_2017-05-18.csv (138f1c72b873...)

Compute resources

Wall time 150.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-18 01:27Z.